

Functional Dependencies in Database Design

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1 Introduction to Functional Dependencies

In the context of **Relational Database Design**, a *Functional Dependency* (FD) describes a precise relationship between attributes in a relation (table). It is a **constraint** stating that certain attributes uniquely determine other attributes.

Definition

A functional dependency $X \rightarrow Y$ between two sets of attributes X and Y in a relation R means:

If two rows (tuples) have the same value for X , then they must also have the same value for Y .

Mathematical Form:

$$\forall t_1, t_2 \in R, t_1[X] = t_2[X] \implies t_1[Y] = t_2[Y]$$

- t_1, t_2 = two different tuples (rows) in the relation.
- $t_1[X]$ = value(s) of attribute(s) X in tuple t_1 .
- $\implies$ = “implies that” in logic.
- Interpretation: *Same X value forces same Y value.*

Example

Consider a **Student** table:

$\text{RollNo} \rightarrow \text{Name, Department}$

This means: Knowing the **RollNo** is enough to know the student's **Name** and **Department**.

RollNo	Name	Department
101	Alice	CS
102	Bob	IT
103	Carol	CS

Real-World Analogy: Think of RollNo like a student's fingerprint — unique for each person. If you know the fingerprint, you know exactly who the student is and what department they belong to.

2 Types of Functional Dependencies — with Examples

Trivial Functional Dependency

A functional dependency $X \rightarrow Y$ is **trivial** if the set Y is a subset of X , i.e., $Y \subseteq X$.

Example:

$$\{\text{Name, RollNo}\} \rightarrow \{\text{Name}\}$$

This FD is trivial and holds by definition; it usually does not provide meaningful design information.

Non-trivial Functional Dependency

If $Y \not\subseteq X$, then the functional dependency $X \rightarrow Y$ is **non-trivial**. These dependencies indicate actual constraints in the data.

Example:

$$\text{RollNo} \rightarrow \text{Name}$$

Fully Functional Dependency

A functional dependency $X \rightarrow Y$ is **fully functional** if no proper subset of X functionally determines Y . This concept is important when X is a composite key (multiple attributes).

Example: Consider relation $\text{Marks}(\text{StudentID}, \text{CourseID}, \text{Marks})$ with FD:

$$\{\text{StudentID}, \text{CourseID}\} \rightarrow \text{Marks}$$

If neither

$$\text{StudentID} \rightarrow \text{Marks} \quad \text{nor} \quad \text{CourseID} \rightarrow \text{Marks}$$

holds, then the FD is **fully functional**.

Partial Dependency

A **partial dependency** exists if an attribute is functionally dependent on part of a composite primary key rather than the whole key. This is undesirable and causes redundancy; it is addressed in the Second Normal Form (2NF).

Example: If $\{\text{StudentID}, \text{CourseID}\}$ is the primary key but:

$$\text{StudentID} \rightarrow \text{StudentName}$$

holds, then `StudentName` is partially dependent on the key.

Transitive Dependency

If $X \rightarrow Y$ and $Y \rightarrow Z$, then by transitivity,

$$X \rightarrow Z$$

Such dependencies cause indirect dependency and are eliminated in Third Normal Form (3NF).

Example:

$$RollNo \rightarrow Dept, \quad Dept \rightarrow DeptHead \quad \implies \quad RollNo \rightarrow DeptHead$$

Multivalued Dependency (MVD) — Brief Note

A **multivalued dependency**, denoted by:

$$X \twoheadrightarrow Y$$

means that for a fixed X , the attributes in Y can take multiple independent values unrelated to other attributes. MVDs are important in higher normal forms like Fourth Normal Form (4NF).

3 Armstrong's Axioms

Armstrong's axioms are a fundamental set of inference rules used to derive all functional dependencies (FDs) that logically follow from a given set of FDs. These axioms are both **sound** and **complete**, meaning all valid dependencies can be inferred using them.

Armstrong's Axioms

1. Reflexivity: If $Y \subseteq X$, then $X \rightarrow Y$. (*A set of attributes always functionally determines its own subsets.*)

2. Augmentation: If $X \rightarrow Y$, then for any Z ,

$$XZ \rightarrow YZ$$

(*Adding attributes to both sides preserves the dependency.*)

3. Transitivity: If $X \rightarrow Y$ and $Y \rightarrow Z$, then

$$X \rightarrow Z$$

(*Dependencies can be chained through intermediate attributes.*)

Detailed Explanation and Examples

Reflexivity Intuitively, if you know the values of attributes in X , you automatically know the values of any subset Y of X . **Example:**

$$\{A, B, C\} \rightarrow \{A, C\}$$

since $\{A, C\} \subseteq \{A, B, C\}$.

Augmentation If X determines Y , then adding any set of attributes Z to both sides preserves the dependency. **Example:** Given $A \rightarrow B$, then for any C ,

$$AC \rightarrow BC$$

also holds.

Transitivity If X determines Y , and Y determines Z , then X determines Z . **Example:** Given:

$$StudentID \rightarrow Dept, \quad Dept \rightarrow DeptHead$$

then:

$$StudentID \rightarrow DeptHead$$

Additional Rules (Derived from Armstrong's Axioms)

Derived Rules

Union If $X \rightarrow Y$ and $X \rightarrow Z$, then:

$$X \rightarrow YZ$$

Decomposition If $X \rightarrow YZ$, then:

$$X \rightarrow Y \quad \text{and} \quad X \rightarrow Z$$

Pseudotransitivity If $X \rightarrow Y$ and $WY \rightarrow Z$, then:

$$WX \rightarrow Z$$

Informal Justifications

- **Reflexivity** is obvious because attributes always determine themselves and any subsets.
- **Augmentation** follows because adding attributes to both sides cannot break the original dependency.
- **Transitivity** works since knowing X lets you find Y , and knowing Y lets you find Z , so X indirectly determines Z .

3.1 Example: Deriving Functional Dependencies Using Armstrong's Axioms

Consider the relation:

$$R = (A, B, C, G, H, I)$$

with the set of functional dependencies:

$$F = \{A \rightarrow B, \quad A \rightarrow C, \quad CG \rightarrow H, \quad CG \rightarrow I, \quad B \rightarrow H\}$$

We want to find some members of the closure F^+ using Armstrong's axioms.

- **Deriving $A \rightarrow H$** Using **transitivity** on:

$$A \rightarrow B \quad \text{and} \quad B \rightarrow H,$$

we conclude:

$$A \rightarrow H.$$

- **Deriving $AG \rightarrow I$** We apply the **augmentation** rule on $A \rightarrow C$ with G :

$$A \rightarrow C \implies AG \rightarrow CG.$$

Then by **transitivity** on

$$AG \rightarrow CG \quad \text{and} \quad CG \rightarrow I,$$

we get:

$$AG \rightarrow I.$$

- **Deriving $CG \rightarrow HI$** Using the **union** rule on:

$$CG \rightarrow H \quad \text{and} \quad CG \rightarrow I,$$

we combine to get:

$$CG \rightarrow HI.$$

4 Real-life Example

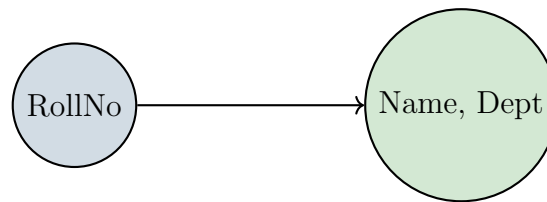
RollNo	Name	Department
101	Alice	CSE
102	Bob	EEE
103	Charlie	CSE

Here:

$$\text{RollNo} \rightarrow \text{Name, Department}$$

This means RollNo uniquely determines both Name and Department.

5 Diagram Representation



Functional Dependencies and Relationship Types

Definition

Functional Dependencies (FDs) can describe how attributes in relations depend on each other. They also indicate the type of relationship between two **entity sets**, such as **Student** and **Instructor**.

1. One-to-One Relationship

Definition

In a **One-to-One** relationship, each student is associated with at most one instructor, and each instructor is associated with at most one student.

Functional Dependencies:

$$\text{Student_ID} \xrightarrow{\text{FD}} \text{Instructor_ID}$$

$$\text{Instructor_ID} \xrightarrow{\text{FD}} \text{Student_ID}$$

Example

Both FDs exist $\Rightarrow$ Each side uniquely determines the other. This mutual determination implies a **1:1 mapping**.

2. Many-to-One Relationship

Definition

In a **Many-to-One** relationship, many students can be assigned to the same instructor, but each student has exactly one instructor.

Functional Dependency:

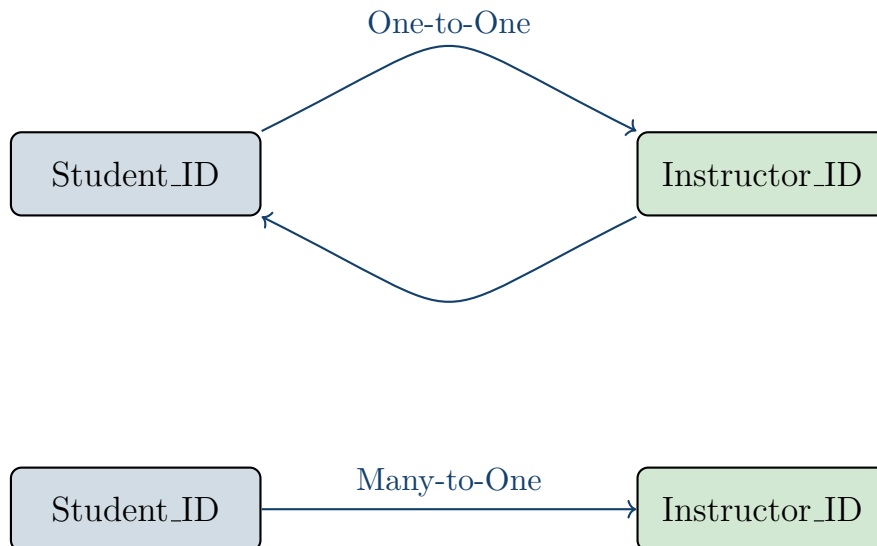
$$\text{Student_ID} \xrightarrow{\text{FD}} \text{Instructor_ID}$$

Example

Only one FD exists from `Student_ID` to `Instructor_ID`. This means:

- Each student is linked to one instructor.
- An instructor can be linked to **multiple** students.

Visual Representation



Functional Dependencies Examples

Example 1

Relation Instance

Consider the following relation (Figure 7.18):

<i>A</i>	<i>B</i>	<i>C</i>
<i>a</i> ₁	<i>b</i> ₁	<i>c</i> ₁
<i>a</i> ₁	<i>b</i> ₁	<i>c</i> ₂
<i>a</i> ₂	<i>b</i> ₁	<i>c</i> ₁
<i>a</i> ₂	<i>b</i> ₁	<i>c</i> ₃

Nontrivial Functional Dependencies

The following nontrivial functional dependencies hold:

$$A \rightarrow B, \quad C \rightarrow B$$

From these, by the union rule, we also get:

$$AC \rightarrow B$$

Why Other Dependencies Do Not Hold

- $C \not\rightarrow A$: tuples 1 and 3 have the same $C = c_1$ but different A values (a_1 and a_2).
- $B \not\rightarrow A$ and $B \not\rightarrow C$: $B = b_1$ appears with multiple A and C values.
- $A \not\rightarrow C$: tuples 1 and 2 have the same $A = a_1$ but different C values (c_1 and c_2).

Example 2

Proof (Union rule is sound using Armstrong's axioms)

Union rule: If $\alpha \rightarrow \beta$ and $\alpha \rightarrow \gamma$ then $\alpha \rightarrow \beta\gamma$.

Proof: Assume

$$(1) \quad \alpha \rightarrow \beta, \quad (2) \quad \alpha \rightarrow \gamma.$$

Apply the *augmentation* axiom to (2) by augmenting with α . This yields

$$\alpha\alpha \rightarrow \alpha\gamma.$$

Interpreting concatenation of attribute sets as set union, $\alpha\alpha = \alpha$. Hence

$$(3) \quad \alpha \rightarrow \alpha\gamma.$$

Now apply the *augmentation* axiom to (1) by augmenting with γ . This gives

$$(4) \quad \alpha\gamma \rightarrow \beta\gamma.$$

Finally, apply the *transitivity* axiom to (3) and (4): from $\alpha \rightarrow \alpha\gamma$ and $\alpha\gamma \rightarrow \beta\gamma$ we conclude

$$\alpha \rightarrow \beta\gamma,$$

which is exactly the union rule. $\square$

Example 3

Proof of Soundness of the Pseudotransitivity Rule Using Armstrong's Axioms

Given:

$$(1) \quad \alpha \rightarrow \beta, \quad (2) \quad \gamma\beta \rightarrow \delta$$

Goal: Prove

$$\alpha\gamma \rightarrow \delta$$

Proof:

1. Apply the *augmentation* axiom on (1) by augmenting with γ :

$$\alpha\gamma \rightarrow \beta\gamma$$

2. By commutativity of attributes in a set (order doesn't matter):

$$\beta\gamma \rightarrow \gamma\beta$$

(This is trivial since attribute sets are unordered.)

3. Using (2), and since $\gamma\beta = \beta\gamma$, we have:

$$\beta\gamma \rightarrow \delta$$

4. Now, apply the *transitivity* axiom to the two FDs from steps 1 and 3:

$$\alpha\gamma \rightarrow \beta\gamma, \quad \beta\gamma \rightarrow \delta \implies \alpha\gamma \rightarrow \delta$$

Hence, the pseudotransitivity rule is sound.

□

Example 4

Testing and Enforcing Functional Dependency $B \rightarrow C$ Using SQL

Consider the relation schema:

$$R = (A, B, C)$$

with relation instance r .

Goal: Test if the functional dependency $B \rightarrow C$ holds on r , and write an SQL assertion to enforce it.

SQL Query to Test $B \rightarrow C$

The functional dependency $B \rightarrow C$ means each value of B determines exactly one value of C .

To test this, find any B value that is associated with more than one distinct C value:

```
SELECT B
FROM r
GROUP BY B
HAVING COUNT(DISTINCT C) > 1;
```

If this query returns any rows, the FD $B \rightarrow C$ is violated.

SQL Assertion to Enforce $B \rightarrow C$

To enforce the FD so that no violations occur, create an assertion:

```
CREATE ASSERTION fd_B_to_C
CHECK (
    NOT EXISTS (
        SELECT B
        FROM r
        GROUP BY B
        HAVING COUNT(DISTINCT C) > 1
    )
);
```

This assertion prevents insertion or update of any tuples violating $B \rightarrow C$.

Note: SQL assertions are part of the SQL standard but are not widely supported in current DBMS implementations. Enforcement in practice often uses triggers or unique constraints.

Example 5

Soundness of Armstrong's Axioms Using the Definition of Functional Dependency

Definition of Functional Dependency

Given a relation schema R and attribute sets $X, Y \subseteq R$, a functional dependency $X \rightarrow Y$ holds if:

For any two tuples $t_1, t_2 \in R$, $t_1[X] = t_2[X] \implies t_1[Y] = t_2[Y]$.

This means the values of X uniquely determine the values of Y .

Armstrong's Axioms and Their Soundness

- **Reflexivity:** If $Y \subseteq X$, then $X \rightarrow Y$.

Proof: Since Y is a subset of X , if two tuples agree on X , they must also agree on Y . Hence, $X \rightarrow Y$ holds.

- **Augmentation:** If $X \rightarrow Y$, then for any Z , $XZ \rightarrow YZ$.

Proof: If two tuples agree on XZ , they agree on X , which implies they agree on Y (by $X \rightarrow Y$). Also, they trivially agree on Z . Thus, they agree on YZ , so $XZ \rightarrow YZ$.

- **Transitivity:** If $X \rightarrow Y$ and $Y \rightarrow Z$, then $X \rightarrow Z$.

Proof: If two tuples agree on X , they agree on Y (since $X \rightarrow Y$), and since they agree on Y , they agree on Z (since $Y \rightarrow Z$). Therefore, $X \rightarrow Z$.

Summary

Axiom	Reason for Soundness
Reflexivity	$Y \subseteq X$ implies agreement on X guarantees agreement on Y .
Augmentation	Extending both sides by Z preserves the dependency since agreement on XZ implies agreement on X and Z .
Transitivity	Dependencies compose: $X \rightarrow Y$ and $Y \rightarrow Z$ imply $X \rightarrow Z$.

6 Conclusion

Functional dependencies are the backbone of relational schema design. They help detect redundancy, guide normalization, and maintain data integrity. Understanding FDs is essential for mastering database design theory.